Where the World Moves: Mapping International Migration Networks

67978 — A Needle in a Data Haystack · Final Project · Group 58

Mohammad Awawda

Domain

International migration is often described one corridor at a time, but that view misses the wider structure behind individual movements. In this project, we model migration as a directed, weighted network: each country is a node, and an edge from country u to country v is the estimated number of people who moved from u to v during a five-year period.

We build six networks, covering 1990–1995 through 2015–2020. Each contains 228–230 countries and roughly 32,000–35,000 positive migration corridors. With the data in this form, we can ask whether the world organizes into migration systems, which countries are structurally central, and how much geography and economics explain before forces such as conflict become visible.

Data

1. Migration flows.
Our main dataset is the bilateral migration-flow dataset of Abel and Cohen, Version 8, obtained from figshare (item 7731233, collection DOI: 10.6084/m9.figshare.c.4470464; methods paper: Scientific Data 6:82, 2019). It is based on UN IMS2024 migrant stocks and WPP2024 demographic data. The CSV is 9.7 MB and contains 313,736 rows and 9 columns: the five-year period (year0), origin (orig), destination (dest), and six different estimates of migration flow.

For the main network, the edge weight is da_pb_closed, the pseudo-Bayesian closed-system estimate. In the checks we ran, the dataset has no missing or negative values; all 1,372 self-loops are exactly zero; and after excluding self-loops, 35.4% of corridor-periods have zero estimated flow. That last point matters later when we choose PPML instead of log-OLS.

The first four periods contain 228^2 country pairs each, while the last two contain 230^2. Country definitions also change over time: Serbia-Montenegro and Sudan appear as the combined codes SCG and SUD before 2010, and are split afterward into SRB/MNE and SDN/SSD. We handle these changes explicitly whenever we compare countries across periods.

These flows are estimates reconstructed from changes in migrant stocks, not direct counts of border crossings. The 1990–1995 total is unusually high partly because the breakup of the USSR and Yugoslavia reclassified some internal migrants as international migrants without physical movement, so we treat the next-period decline mainly as a return toward baseline (Fig. 1). Since the estimator is calibrated mainly against reported European flows, we focus on network structure rather than treating every absolute value as exact.

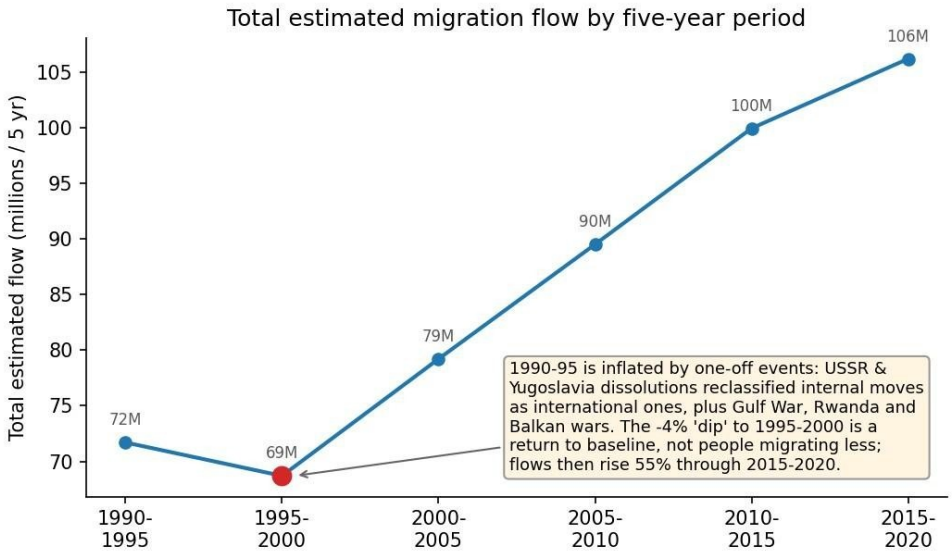


Fig. 1. Total estimated migration flow by five-year period. The unusually high 1990–1995 estimate is partly driven by the breakup of the USSR and Yugoslavia, which reclassified some internal migrants as international migrants. The following decline should therefore be read mainly as a return toward baseline rather than a sudden collapse in migration.

Some individual corridors also contain estimation artifacts. The clearest example in our data is the estimated USA→Cambodia flow of about 431,000 people in 1990–1995, which is larger than the Cambodian-born population living in the United States at the time. We keep it as a sanity check rather than removing it quietly, and later test whether the hub rankings survive under a different migration-flow estimator.

2. Gravity covariates.

To study how much migration can be explained by economic and geographic factors, we use the CEPII Gravity database, version V202211. For our analysis we retain the six relevant years and 20 fields. These include distance, contiguity, common official language, colonial relationships, common colonizer, and the GDP and population of both the origin and destination countries.

CEPII sometimes has multiple country IDs sharing one ISO3 code, including mostly empty records for entities outside their lifetimes. We keep the populated row for each (year, origin, destination), verify that no dyad-year has two competing populated rows, and obtain 354,294 unique dyad-years; the later merge is checked to be one-to-one.

Coverage is not uniform. Some small territories and lower-data countries are missing important CEPII covariates, so the gravity model cannot use every migration corridor. We therefore interpret the gravity results as applying to the available-covariate sample rather than assuming they automatically describe the entire migration network.

3. Conflict data.

For the conflict analysis we use the UCDP/PRIO Armed Conflict Dataset, version 26.1. It is a 0.52 MB CSV containing 2,816 conflict-years and 28 columns, covering 1946–2025. When a conflict location contains several countries, we split the location and map each country to its ISO3 code.

For our analysis, a country-period is marked as being in war if at least one UCDP conflict-year with intensity level 2 — at least 1,000 battle deaths in a year — occurs within that five-year migration period. We also handle the Sudan and Serbia-Montenegro succession cases so that these conflict labels line up correctly with the migration networks.

4. Basemap.

For the geographic visualization we use Natural Earth 110m country borders, a roughly 0.8 MB GeoJSON containing 177 features. The basemap is used only for visualization; it does not determine which countries are included in the network analysis. Because it is a low-resolution world map, some very small countries and territories do not appear on it even though they are still present in the migration network.

The migration flows come from figshare, the gravity variables from CEPII, the conflict data from UCDP, and the basemap from Natural Earth. Exact dataset versions and the full provenance trail are included in the project's data/PROVENANCE.txt file and README.

Problem 1 — Community structure: migration systems, not continents

Problem description. We test whether migration flows organize countries into meaningful communities (H1), whether these communities reflect institutions and historical links rather than simply continents (H2), and whether their strength and membership change from 1990 to 2020 (H3).

Solution. For each period, we build a directed, weighted graph and run Infomap with num_trials=20, seed=42, and teleportation probability 0.15. Infomap fits the problem because it follows directed flow: if a random walker tends to stay inside a group of countries before leaving, that group behaves like a natural community.

We compare this with Louvain on the same directed, weighted graphs using NetworkX directed modularity (seed=42). We track modules across periods by greedily matching consecutive communities with Jaccard overlap ≥ 0.2, while accounting for Sudan and Serbia-Montenegro succession chains.

We also report Infomap compression savings, $1 - L(M)/L(1)$, where larger values mean the detected communities describe migration flow more efficiently than a single unstructured network.

Evaluation criteria. We compare detected communities with six integration or free-movement blocs — EU-28, GCC, ECOWAS, the Mercosur core, the post-Soviet space, and ASEAN — covering 75 countries, using NMI and ARI. Baselines are the continent partition and 1,000 random permutations that preserve community sizes.

Bloc membership is fixed around 2015, so early-period scores show how visible those later-associated structures already were, not that the institutions existed in the same form in 1990. We also check whether the structure could come from degree alone, that is, from how many corridors each country has rather than from which ones.

Setup. We use two null models.

Null A — degrees kept, wiring random — generates 100 networks per period by directed edge swaps: corridors $u \to v$ and $x \to y$ become $u \to y$ and $x \to v$ when this creates no self-loop or duplicate, repeated twice per edge. Every country keeps exactly its number of origins and destinations; who it connects to is random. The real weights are then dealt out at random over the rewired corridors. This is the configuration-model null from class: degree heterogeneity stays, geography, history, and volume structure go. Infomap runs with the same 20 trials as on the real networks.

Null B — topology kept — applies only to 2015: we keep the real set of corridors, randomly shuffle their weights across edges, generate 100 networks, and rerun Infomap to separate the effect of corridor existence from the distribution of migration volume.

Results. Infomap finds 14–18 communities per period, and many of them correspond to recognizable migration systems rather than simple continental divisions.

The maps show a Gulf–South Asia labor system, France with the Maghreb, and a post-Soviet system that includes Israel.

Infomap's NMI against the six blocs is 0.694, 0.669, 0.683, 0.690, 0.785, and 0.778 from 1990 through 2015, and its ARI is 0.49, 0.40, 0.36, 0.43, 0.62, and 0.62. The continent partition scores 0.745 (NMI) and 0.656 (ARI), so Infomap does not beat simple geography in the first four periods, but does in 2010 and 2015. Random permutations of the detected labels score far lower (mean NMI 0.16–0.19, 95th percentile 0.21–0.24), so both partitions carry real information about the blocs. Later communities therefore align more strongly with institutional and historical blocs, not throughout the full 30 years.

At the same time, compression savings fall from 0.178 in 1990 to 0.118 in 2015. The communities become easier to name, but the global network is less tightly trapped inside regional systems.

Louvain produces fewer, coarser communities (7–10 per period), with modularity falling from roughly 0.60 to 0.47. On the bloc benchmark the two are close, and Louvain is not the weaker one: its NMI is slightly higher in 1990–2000 (0.709 vs. 0.694, 0.690 vs. 0.669, 0.686 vs. 0.683) and its ARI is higher in five of the six periods (0.542 vs. 0.397 in 1995, 0.623 vs. 0.617 in 2015). Infomap is ahead only on NMI in 2010 and 2015 (0.785 vs. 0.697, 0.778 vs. 0.741). We keep Infomap as the main method because it follows directed flow and resolves 14–18 systems instead of 7–10, not because it scores better.

The null models give the strongest check on this result. Null A shows how much structure degrees alone buy: the rewired networks reach compression savings of 0.097 on average in 1990 and 0.064 in 2015, with NMI around 0.34.

The real networks beat all 100 replicates in every period on both measures (empirical $p < 0.01$), and what is left after subtracting the null mean falls from 0.081 in 1990 to 0.054 in 2015, so the decline behind H3 is not just the network gaining corridors over time.

For Null B in 2015, compression savings average about 0.066 and never exceed 0.086, while NMI averages 0.347 and never exceeds 0.401. The real network reaches 0.118 and 0.778, and all 100 shuffled-weight networks fall below it (empirical $p < 0.01$). Thus, the 2015 structure depends on migration volumes, not only on which corridors exist.

Visualization. Fig. 2 follows the detected communities across all six periods, making it possible to see which systems persist, merge, or change membership. Fig. 3 shows the 2015 communities geographically, while Fig. 4 places the observed 2015 community strength against both null distributions and tracks observed against null savings in every period. Together, the three views move from what the communities look like, to where they are, to whether the structure is stronger than chance-based alternatives.

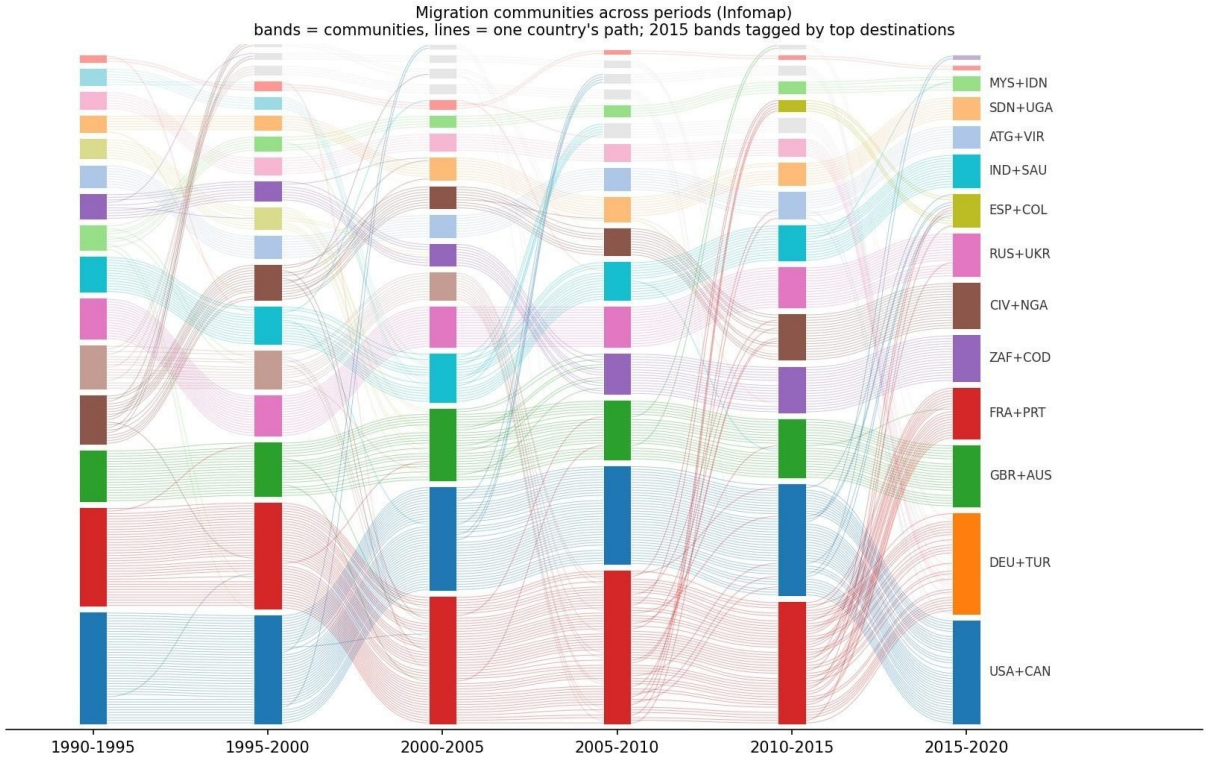


Fig. 2. Migration communities across all six periods. Each band represents an Infomap community and each line follows one country through time. Stable and changing migration systems can be seen directly, with the 2015 communities labeled by their largest destinations.

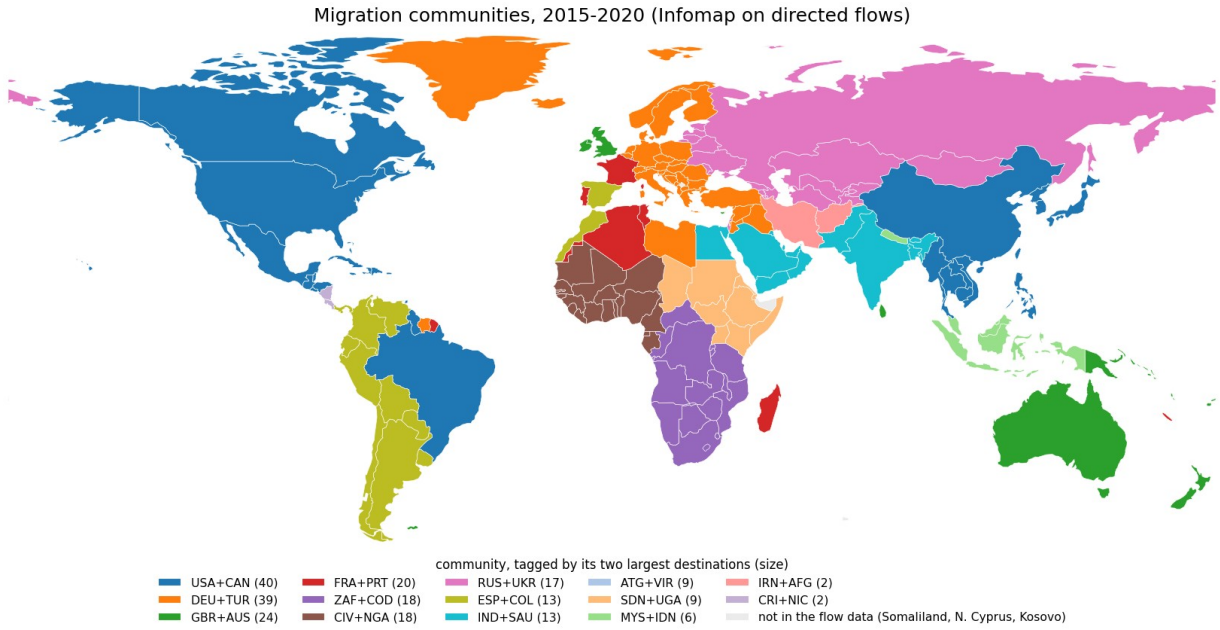


Fig. 3. Migration communities in 2015–2020. The geographic view highlights systems that cross continental boundaries, including Spain with parts of Hispanic America and the Gulf with South Asia. Legend tags are the community's two largest destinations, the same tags as the 2015 bands in Fig. 2. 58 of the 230 network countries are too small to show on the 110m basemap; they are still in the analysis.

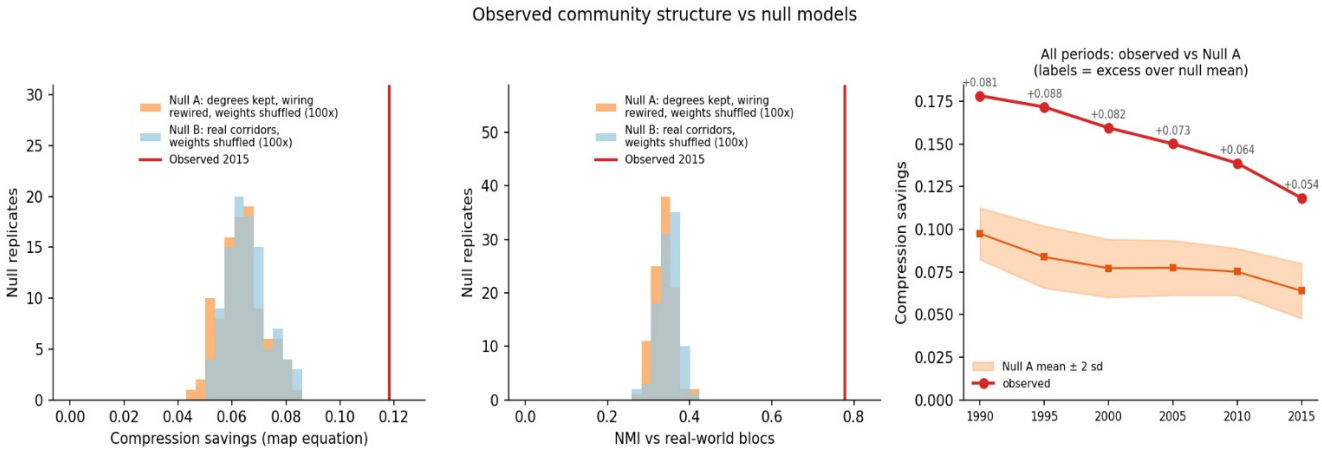


Fig. 4. Observed community structure compared with both null models. Left and middle: 2015. Null A keeps every country's number of corridors and randomizes everything else; Null B keeps the real corridors and shuffles their volumes. Both stay well below the observed compression savings and NMI. Right: observed versus Null A savings in every period; the labels give the excess over the null mean, which falls from 0.081 to 0.054. The structure is real in every period, and its decline is not explained by the growing number of corridors.

Impediments. There is no perfect ground truth for a “true” migration community. The bloc benchmark is useful but partly anachronistic for early periods, and choices such as keeping the Mercosur benchmark to its core members affect interpretation. Our first Null A was a size-only multinomial network: nearly complete, with weights that are just the product of origin and destination totals, and Infomap never found more than one module in it. A null that cannot produce structure cannot tell us whether ours is unusual, so we replaced it with the rewiring null above.

Problem 2 — Hubs: who is central, and how has it changed?

Problem description. The largest migration destination is not necessarily the most structurally important country in the network. One country may receive many migrants from a small set of origins, while another may sit at the center of several important migration pathways.

Our second problem therefore asks: **Which countries are the structural hubs of global migration, how has that changed between 1990 and 2020, and what can network centrality reveal that raw migration volume cannot?**

Solution. For each period, we measure centrality from several angles.

Our main measure is **weighted PageRank**, with α = 0.85. PageRank follows migration flows and gives high scores to countries receiving flow from origins that are themselves central, so **where the flow comes from also matters**.

The damping parameter α controls how often the walker follows the network instead of teleporting. We repeat the analysis with α = 0.80 and α = 0.90 to check that rankings are not driven by this choice.

We also calculate **in-strength**, **out-strength**, and **weighted betweenness centrality**. For betweenness, edge distance is 1/flow, so heavily traveled corridors are short connections. Since this is only one way to convert flow into distance, we interpret betweenness qualitatively.

When following rankings through time, we also use the same state-succession chains as before so that SUD→SDN and SCG→SRB can be tracked consistently.

Evaluation criteria and setup. There is no objective ground-truth label for "the world's most structurally important migration countries," so instead of pretending that one exists, we evaluate whether the result is **robust**.

We use three checks.

First, **damping sensitivity**: changing α from 0.85 to nearby values should not completely reorder the countries.

Second, **estimator sensitivity**: our main networks use the `da_pb_closed` migration estimator, but the underlying migration data contain several alternative estimates. We rebuild the 1990 and 2015 networks using `sd_drop_neg`, which is based on a substantially different stock-differencing method, and compare both the overall PageRank rankings and their top-10 countries.

Third, we compare the 2015 PageRank top-10 with the **UN International Migrant Stock top-10 destinations** around 2020. This is only a consistency check, not independent validation, because our estimated migration flows are themselves partly constructed from UN migrant-stock data.

Since there is no true PageRank ranking to compare against, we focus on whether the main hub findings survive reasonable changes in how we construct and parameterize the network.

Results. The United States is the clearest constant: it ranks **first in PageRank in every period**. Beyond that, however, the hub structure changes considerably.

The 1990 top-10 is: **USA, Germany, France, Cambodia, Canada, Australia, UK, Mexico, Italy, Russia.**

By 2015 it becomes: **USA, Russia, Germany, Ukraine, France, Spain, UK, Turkey, Mexico, Australia.**

The largest changes are Russia rising from **10th to 2nd**, Ukraine from **22nd to 4th**, and Turkey from **20th to 8th**, while Canada falls from **5th to 12th**. These shifts show that network position and raw migration volume are not the same thing.

PageRank is extremely stable across damping values: rankings at α = 0.85 have Spearman correlation of at least **0.997** with α = 0.80 and α = 0.90 in every period.

The UN stock comparison gives a **7/10 overlap** in 2015. PageRank excludes Saudi Arabia, the UAE, and Canada but includes Mexico, Turkey, and Ukraine, showing how central origins and regional systems can matter beyond migrant-stock size.

Betweenness adds a different view: countries such as **India, Saudi Arabia, Bangladesh, and South Sudan** can play bridging roles even when they are not near the top by PageRank.

The alternative-estimator check is more cautious. The PageRank **top-10 overlap is 7/10 in 1990 and 6/10 in 2015**, but the Spearman correlation across the full ranking is only **0.55 and 0.49**, respectively. The exact ordering of the long tail depends considerably on how migration flows are estimated, so we claim robustness for the **main hubs near the top of the ranking**, not for every country's precise rank.

Visualization. Fig. 5 shows how PageRank rankings changed between 1990 and 2015. The slope chart emphasizes the largest movements rather than forcing the reader to compare two long ranked lists, making the rise of countries such as Russia, Ukraine, Turkey, and Kazakhstan particularly visible.

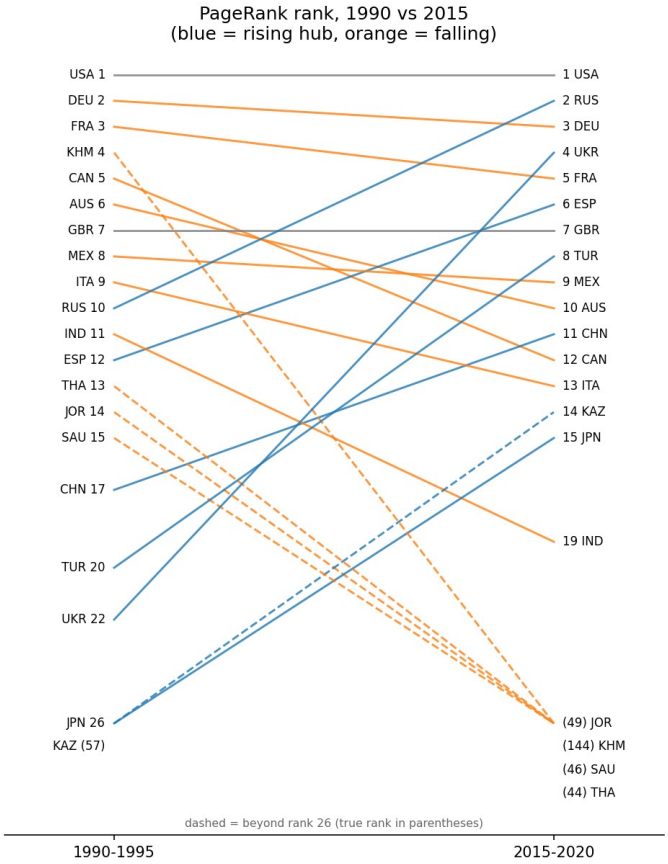


Fig. 5. PageRank rank changes from 1990 to 2015. Blue lines indicate countries rising in structural importance and orange lines indicate countries falling. Dashed lines mark countries whose true rank lies beyond the displayed top 26. Cambodia is retained as the main estimator-artifact case study.

Impediments. Cambodia's 1990 PageRank rank 4 is the clearest warning case. The primary estimator assigns it about **1.30 million incoming migrants**, including roughly **431,000 from the United States**, larger than the Cambodian-born population then living in the United States.

We did **not** delete Cambodia. Under the alternative `sd_drop_neg` estimator, it falls from **PageRank rank 4 to rank 130**, confirming that this hub result is highly estimator-dependent and that the top hubs are more reliable than every exact rank.

Problem 3 — Gravity: what economics explains, and what it misses

Problem description. Our final problem asks how far a fairly standard economic model can take us before the migration network reveals patterns that economics alone does not explain.

We test three related hypotheses. H4: a small set of geographic and economic variables explains a substantial part of the variation between migration corridors. H5: some of the model's largest misses correspond to politically disrupted or displacement-related corridors. H6: countries experiencing war tend to send more migrants than the gravity model predicts.

Solution. We fit a gravity model, using the same intuition as Newton's law: movement tends to increase with the size of the two countries and decrease with the distance between them. The full model includes log distance, log GDP and log population for both origin and destination, contiguity, common official language, colonial ties, common colonizer, and period fixed effects.

PPML is a better fit than log-OLS here because 35.4% of non-self corridor-periods have zero estimated flow, and migration flows are strongly heteroskedastic. PPML keeps the zeros and estimates the model directly in levels.

The full model is compared with a distance-plus-period baseline. We cluster standard errors by origin-destination pair and use the UCDP war flags defined in the Data section, where a country-period is marked as war if an intensity-level-2 conflict occurs within the five-year window.

Evaluation criteria and setup. We use a fixed random 80/20 train-test split with seed 42, giving 165,885 training rows and 41,437 test rows. We compare models using out-of-sample Spearman correlation, Pearson correlation on log-transformed flows, and in-sample deviance pseudo-R^2.

For H5, we inspect corridors with at least 50,000 migrants and rank them by actual/predicted migration. For H6, we aggregate migration by origin-period and compare war and peace origins with a one-sided Mann–Whitney U test because the residual distribution is skewed and contains extreme values.

The CEPII merge is checked to remain one-to-one. Of the 312,364 non-self migration corridor-periods, 207,322 have all covariates required by the gravity model. Missingness is concentrated in small territories and lower-data countries, so the gravity and conflict results should be interpreted as applying to the available-covariate sample rather than automatically to every country in the network.

We define a corridor residual as log(1+actual) − log(1+predicted), and also report the ratio (actual+1)/(predicted+1), which is easier to interpret directly.

Variable	Coef.	Clust. SE	Variable	Coef.	Clust. SE
ln distance	-0.656	0.075	contiguity	+1.308	0.273
ln GDP dest	+0.504	0.051	common off. language	+0.601	0.142
ln GDP origin	+0.002	0.034	colonial tie (ever)	+1.023	0.146
ln pop origin	+0.626	0.041	common colonizer	+0.760	0.193
ln pop dest	+0.150	0.044	period fixed effects	yes	—

Results. The coefficients behave broadly as expected: distance is negative (about −0.66), while contiguity (+1.31), colonial ties (+1.02), common official language (+0.60), destination GDP (+0.50), and origin population are positive. Origin GDP is essentially zero once the other variables are included.

The full model improves clearly on distance alone: deviance pseudo-R^2 rises from 0.129 to 0.607; test Spearman from 0.335 to 0.623; and Pearson-on-logs from 0.419 to 0.644. It also predicts total scale closely, about 451 million migrants versus 454 million observed. H4 is supported, although plenty remains unexplained.

Looking at the largest residuals revealed patterns the overall fit could not show. Several large underpredictions correspond to conflict, displacement, or return migration: Guinea→Liberia in 1995 is about 822 times predicted, Bhutan→Nepal in 1990 about 529 times, and Malawi→Mozambique in 1990 about 390 times. The known USA→Cambodia artifact also appears, which reminded us that residuals can flag either real events or data problems.

The overpredictions are useful too. The model predicts around 1.67 million migrants from India to Pakistan in 2015–2020, but the estimated flow is only about 224,000, or roughly 13% of the prediction. In our data, this is a clear example of political barriers suppressing movement between large neighboring countries.

H5 is therefore supported cautiously: many large residuals occur where conflict, political barriers, return migration, or data artifacts introduce forces missing from the economic specification.

For H6, war-associated origins have higher residual outflow: median origin-level log ratio −0.079 for war origins versus −0.349 for peace origins, with one-sided Mann–Whitney $p = 0.012$ across 82 war and 1,035 peace origin-periods. A largest-corridor metric points the same way but is not significant ($p = 0.17$).

We interpret this as an association, not evidence that war causally produces a particular amount of migration. The conflict window overlaps the migration window, and the model does not include every factor that differs between conflict and non-conflict countries.

Visualization. Fig. 6 compares predicted and actual migration and highlights underpredicted large corridors. Fig. 7 summarizes origin-level residual outflow for war and peace periods.

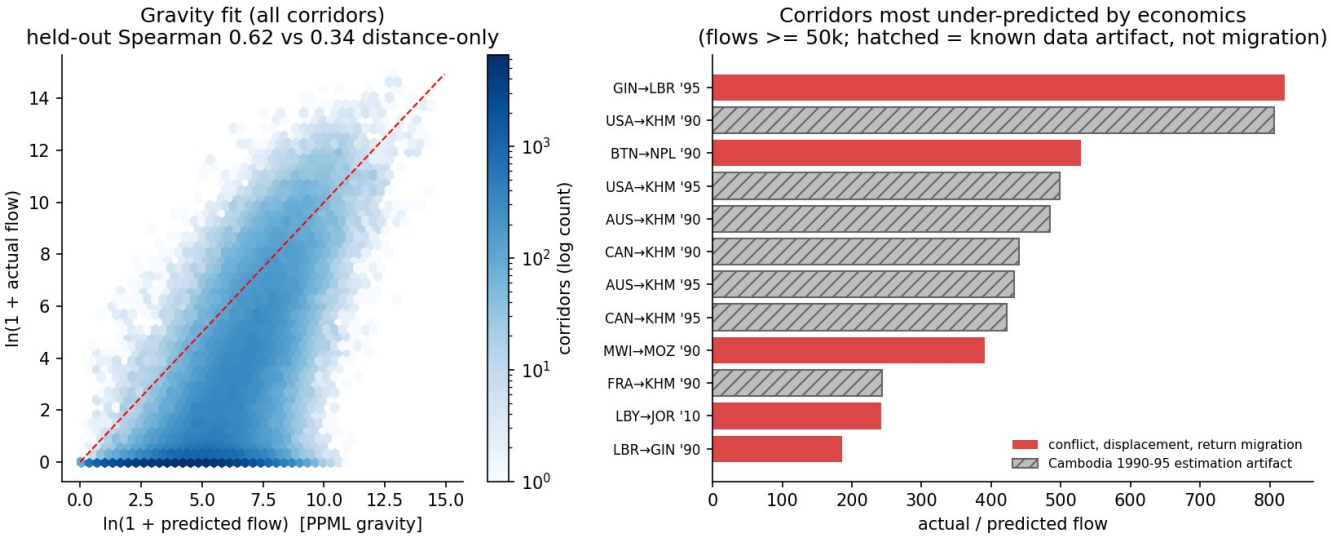


Fig. 6. Gravity-model fit and the largest underpredicted corridors. The left panel compares predicted and observed migration, while the right panel shows large corridors with the greatest actual-to-predicted ratios. Red bars are conflict, displacement, or return migration; the hatched gray bars are the 1990–95 flows into Cambodia, which we treat as an estimation artifact (see Problem 2), not as migration the model missed.

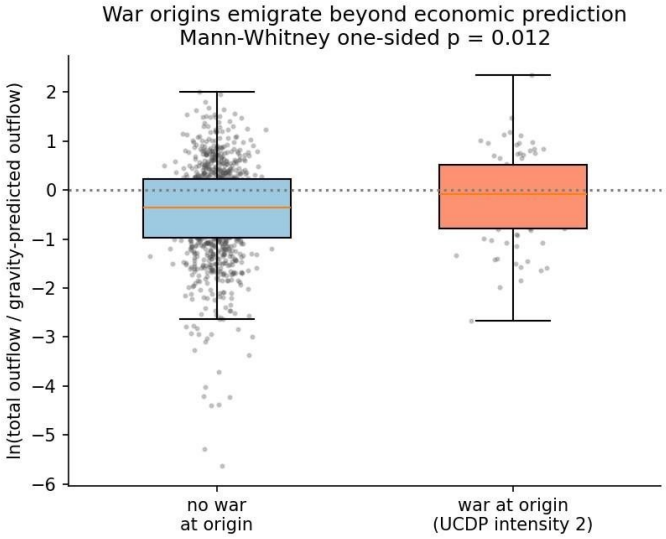


Fig. 7. Gravity residual outflow for war and non-war origins. War-associated origin-periods tend to have higher residual outflow than peace origin-periods. The result is statistically significant in the main analysis, but it should be interpreted as an association rather than a causal estimate of displacement.

Impediments. Three data issues turned out to matter. First, our original H6 mean-corridor residual metric failed (p = 1.0) because war migration can concentrate into a few large corridors while many destinations remain near zero; aggregating total outflow gave a more meaningful test.

Second, the first CEPII merge increased row counts because duplicate ISO3 identities existed around state transitions; keeping the populated dyad-year record and asserting one-to-one merges fixed it. Third, UCDP successor-state codes initially failed to match pre-2010 SUD and SCG, and mapping them back changed the H6 p-value only slightly, from about 0.015 to 0.012.

Finally, the gravity specification is deliberately simple. Without origin-period and destination-period fixed effects, coefficients are descriptive rather than causal; the random split can place the same dyad in both train and test; and because conflict overlaps the migration window, H6 is only an association, not a causal estimate of displacement.

Future Work

Next steps would be to strengthen the gravity model with origin-period and destination-period fixed effects and leave-one-period-out validation; use Abel and Cohen companion estimates for outward, return, transit, and sex-specific flows; apply temporal or multilayer Infomap; and add UNHCR refugee data to separate conflict-driven displacement from labor and family migration.

Conclusion

Viewing migration as a network lets us see structure that corridor-by-corridor analysis can miss: later-period communities increasingly align with institutional and historical blocs, even as overall compression savings decline and the network becomes less regionally trapped. The hub analysis shows that structural importance is not raw volume. PageRank highlights countries whose importance depends on where flows come from, while the Cambodia case and alternative-estimator check show why broad hub patterns are more reliable than every precise rank.

The gravity model explains a substantial share of corridor structure, but its residuals show where economics and geography are incomplete: conflict, political barriers, return migration, and data artifacts all appear in the largest deviations. The reality checks — including the dissolution effect, state-succession fixes, and failed residual metric — became part of the analysis rather than cleanup afterward.